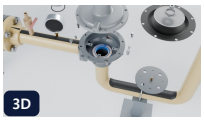
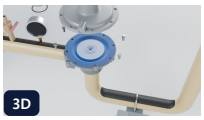
Fisher™ 289H relief valve, 2 in. — Step-by-step

One page to follow along with the ② disassembly & assembly guide video — the ▶ time on the right opens that part of the video.

② Disassembly guide
English

Work by qualified personnel following the manufacturer's manual · KFNGS training material — not an Emerson publication
Work the manual does not describe: do not proceed until the manufacturer's procedure is confirmed
torques from the manual only · tool sizes from KFNGS service records — verify the actual fastener

Photo	Step · what to do / see	Tools · torque	★ Caution · check	Video
 3D	**Before you start — isolation and pressure release** First decide how the downstream system is protected during isolation, then close the upstream block valve and release all trapped pressure to a safe place (pp. 4, 6).	—	The spring (7) is compressed on the diaphragm — release the compression first (p. 7).	▶ 0:07
 Real 2024	**① Closing cap · relieve spring compression** Remove the closing cap (14) and gasket (15), then turn the adjusting screw (6) counterclockwise to relieve spring compression (p. 7, step 1).	—	The manual gives no amount or end point · a new gasket is recommended each time the cap is opened (p. 3).	▶ 1:04
 Real 2024	**② Lift off the spring case** The manual does not describe how to confirm that spring compression has been fully released — do not proceed until the manufacturer's procedure is confirmed. Then remove the machine screws (8) and the spring case (2) per p. 7, step 2.	Machine screws: 7/16 in. spanner or socket **Torque: check the manual value**	The drive screw in the case stops the set pressure being raised too high (p. 4).	▶ 1:23
 3D	**③ Adjusting screw · washer · spring** Unscrew the adjusting screw (6) out through the top of the case and remove the washer (27) and spring (7) (p. 7, step 2).	—	Match the spring colour and drive-screw position against Tables 1 and 2.	▶ 1:46
 Real 2024	**④ Diaphragm assembly** If fitted, remove the lifting-stem nut (24) and stem (25) first. Remove the hex nut (11), then the lower spring seat, diaphragm head, diaphragm, lower head and gasket (p. 7, steps 3–4).	Hex nut: 7/16 in. **Torque: check the manual value**	The diaphragm and gasket are kit items — replace if damaged (p. 8).	▶ 2:06
 3D	**⑤ Stem guide assembly** Remove the 4 machine screws (29) crisscross and lift the guide assembly straight out with its parts (p. 7, step 5).	Machine screws: screwdriver **Torque: check the manual value**	The seat O-ring (20) and orifice (10) stay in the body.	▶ 2:28
 Real 2024	**⑥⑦ Guide · pitot-tube parts · orifice** Slide the guide off the pitot tube and separate the spacer, gaskets, O-ring washer, seat O-ring and orifice (p. 7, steps 6–7).	—	★ Clear debris from the pitot-tube tip and guide-plate holes; check the O-ring washer seating face.	▶ 2:47
 3D	**Outlet · vent piping check** Check that the vent piping and its end (rain cap, elbow) are not blocked (pp. 2–4).	—	The vent end must be in a safe, well-ventilated place away from buildings and windows (p. 3).	▶ 3:58
 3D	**Assembly (reverse order)** The manual text and drawing give different materials for the orifice (10) threads — do not reinstall the orifice until the manufacturer's procedure is confirmed. For the remaining parts, assemble in reverse order — the diaphragm convolution must face down (Fig. 3); tighten screws evenly crisscross.	**Torque: no values in the manual — ask the manufacturer**	★ When changing the spring range, use a new spring case (p. 7 note).	▶ 4:17
 3D	**Pre-pressurizing checks · commissioning** Check the vent piping and spring range; confirm the spring-case vent is plugged or undrilled (p. 2) → close the depressurization vent valve; monitor inlet pressure while slowly opening the upstream block valve and setting (clockwise = higher) → new gasket and closing cap (pp. 3, 7). If a discharge-line valve is fitted, confirm it is locked open per site procedure before pressurizing.	—	The manual has no set-point test method — do not proceed until the manufacturer's procedure is confirmed.	▶ 6:14

Photos: Real = KFNGS service records (year · 299HS = parts shared with the 299H) · 3D = KFNGS guide screen (training reconstruction) · ①… = guide video steps

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Installation, operation and maintenance must be performed by qualified personnel following applicable codes, site safety procedures and the manufacturer's manual.
The disassembly & assembly guides are training reconstructions, not manufacturer procedures. Do not perform any work the manual does not describe, including releasing spring force, until you have checked the manufacturer's procedure.
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